

Digital Export Enablement Platform via Dak Ghar Niryat Kendra (DNK) • India Post

Executive Mission: **Niryat Saathi** empowers India's rural artisans, weavers, SHGs, and MSMEs to access international trade directly through their local **Dak Ghar Niryat Kendra (DNK)** post office. The platform automates the entire export lifecycle: from multilingual AI compliance advisory and automated Postal Bill of Export (PBE-III/IV) generation to counter verification, digital customs dispatch via India Post EMS, and global parcel tracking.

All services run synchronously and connect through real-time REST and WebSocket telemetry:

Pre-configured 1-click accounts seeded directly in MongoDB for instant evaluation:

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3. Complete End-to-End System Project Flow

The platform connects local rural production with international global delivery across 8 distinct stages:

- **Phase 1: Seller Onboarding & Export Readiness Scoring:** Artisans and SHGs register on the Seller Portal. The system analyzes business registration (Udyam / GST), product photos, and material declarations, calculating a dynamic **Export Readiness Score (0-100%)** with actionable recommendations to unlock international trade.
- **Phase 2: Daksh Multilingual AI Export Consultation:** Sellers consult **Daksh**, an AI assistant powered by Groq LLM. Daksh speaks **English, Hindi, and Gujarati**, advising on destination import duties, HS Codes, mandatory certificates (Phytosanitary, CoO, IEC exemptions for postal exports), and packaging standards.
- **Phase 3: Product Cataloging & Global Marketplace Exposure:** Sellers upload handicrafts, textiles, and organic goods with volumetric weight, dimensions, and HS codes. The products immediately appear on the **Global Buyer Marketplace** with real-time currency conversion (INR to USD/EUR) and Request-for-Quote (RFQ) triggers.
- **Phase 4: Order Placement & Automated Customs Documentation:** When an international buyer confirms an order, Niryat Saathi's backend compliance engine automatically drafts the required customs documentation: **Commercial Invoice, Packing List, and Certificate of Origin (CoO)**.
- **Phase 5: Digital Postal Bill of Export (PBE) Generation:** In compliance with CBIC and India Post regulations, the platform auto-generates either **PBE-III (Postal Export for gifts/samples)** or **PBE-IV (Commercial Postal Export)** complete with unique tracking barcodes, HS code manifests, and consignment values.
- **Phase 6: DNK Counter Verification & Acceptance:** The seller brings the pre-booked parcel to their nearest **Dak Ghar Niryat Kendra (DNK)**. The postal operator logs into **/operator**, scans the parcel's digital barcode, verifies physical dimensions/weight, approves the customs declaration, and generates the India Post EMS receipt.
- **Phase 7: India Post International EMS Dispatch & Tracking:** The parcel is handed over to India Post's International EMS network. Both buyer and seller receive live status updates via automated notifications and visual timeline tracking.
- **Phase 8: National Postal Admin Intelligence & Diagnostics:** Postal authorities on **/admin** view real-time macro analytics: total export volume (■■), top exported categories (Textiles, Handicrafts, Spices), active DNK performance rankings, and live MongoDB telemetry.

4. Full-Stack Architecture & Technology Components

Technology Layer	Frameworks & Tools	Key Architectural Value
Web Frontend	React 19, TypeScript, Vite 8, Tailwind CSS v4, Zustand 5, Lucide Icons, Recharts	Role-based navigation, zero-latency state sync, interactive charts, printable PDF/HTML exports, multilingual localization (EN, HI, GU).
Mobile Application	React Native, Expo 54, Expo Metro Bundler, Vector Icons, Safe Area Context	Cross-platform accessibility designed for field sellers and DNK counter operators on Android/iOS/Web.
Backend API Gateway	Python 3.14, Django 6, Django REST Framework (DRF), PyMongo, StatReloader	High-throughput RESTful endpoints, CORS security headers, token-based authentication, and automated backup daemon.
Database Engine	MongoDB 8.x (NoSQL Document Store), PyMongo Singleton Pool	High-flexibility JSON documents allowing schema evolution for international postal regulations without database locks.
Generative AI Engine	Groq Cloud API, OpenAI/Llama Model, Prompt Heuristics	Sub-second AI inference advising rural exporters in regional Indian vernacular languages with regulatory accuracy.

5. MongoDB Database Collections Schema (sih_dakghar_db)

14 dedicated collections containing 173 seeded documents for zero-mock, end-to-end functionality:

Collection Name	Docs	Core Schema Attributes	System Role / Relationship
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users	4	id, name, email, password, role, phone	Auth credentials for Seller, Operator, Admin, Buyer
sellers	10	id, userId, businessName, businessType, readiness	MSME profile, state, pincode, export score
products	22	id, sellerId, name, price, hsCode, weight, stock	Export catalog items with dimensions & materials
orders	20	id, buyerId, sellerId, productId, amount, status	International buyer orders and purchase lifecycles
shipments	10	id, orderId, trackingNumber, dnkId, destination	India Post EMS tracking records & logistics stages
dnks	20	id, name, postOfficeCode, address, city, state	Physical Dak Ghar Niryat Kendra post offices
documents	30	id, sellerId, orderId, type, docNumber, status	PBE-III, PBE-IV, Invoices, Certificates of Origin
compliance_rules	12	id, productCategory, destination, requirement	Country-specific tariffs, duties, and restrictions
assistant_qa	8	question, answer, nextSteps, requiredDocuments	Pre-indexed knowledge base for export queries
export_journey_steps	9	step, title, description, responsibleParty	Configurable 9-step export progression timeline
notifications	20	id, userId, title, message, type, readStatus	Cross-role operational alerts & order status pings
support_tickets	6	id, userId, orderId, category, priority, status	Helpdesk tickets for customs clearance & logistics
feedback	2	userName, userEmail, rating, category, message	User ratings, satisfaction scores & UI suggestions

6. Project Execution & 1-Click Launch Guide

The entire platform can be launched in a single command using the pre-configured Windows batch script or individually:

Execution Mode	Terminal Command	What it Does
1-Click Master Launch (Recommended)	<code>.\run_project.bat</code>	Checks MongoDB, seeds all 173 records, launches Django Backend (8000), Vite Frontend (5173), and Expo Mobile (8081) in dedicated consoles.
Backend API & Console	<code>cd backend .\venv\Scripts\python.exe manage.py runserver 127.0.0.1:8000</code>	Starts Django REST Framework server and single-page developer console.
Web Frontend (Vite)	<code>cd Dakh-main npm run dev -- --host 127.0.0.1 --port 5173</code>	Starts Vite development server for Seller, Operator, Admin, and Buyer portals.
Mobile App (Expo)	<code>cd mobile npx expo start --web --port 8081</code>	Starts Expo Metro Bundler and cross-platform web simulator on port 8081.
Test & Seed DB (CLI)	<code>cd backend .\venv\Scripts\python.exe test_backend_and_mongo.py</code>	Executes automated health checks against MongoDB and all 16 REST endpoints.

Summary & SIH 2026 Impact: Niryat Saathi delivers an institutional-grade, production-ready solution that bridges local Indian rural production with global buyers. By integrating India Post's unmatched postal footprint with intelligent software, the platform turns every local post office into a global export gateway.